SOON YOUNG KWON

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EDUCATION

HARVEY MUDD COLLEGE (HMC) Claremont, CA – Bachelor of Engineering — Class of 2027

Major in General Engineering; Specialization in Mechanical Engineering and Material Science and Engineering

Relevant Coursework: Multivariable Calculus, Linear Algebra, CS, Chemistry, Mechanics, Engineering Projects (E4),

SUZHOU SINGAPORE INTERNATIONAL SCHOOL Suzhou, Jiangsu, China - IB Diploma — Graduated May 2023

Physics Higher Level, Mathematics Analysis and Approaches HL, Economics HL — Final Unweighted GPA: 3.92

PROJECTS & EXPERIENCE

E4 ENGINEERING PROJECT — January 2024 – May 2024

- Designed and prototyped a universal antenna remounting system for satellite/drone GPS tracking, enabling seamless integration with a motorized telescope mount while ensuring ±0.1mm positional accuracy via GD&T-compliant tolerancing
- Developed antenna brackets using SolidWorks, optimizing for lightweight (CFRP 3D printing) and high-strength (CNC-machined aluminum) components
- Managed project timeline and coordination using Gantt charts, daily stand-ups, and meeting minutes for on-time delivery

MUDD AUTOMOTIVE CLUB — September 2023 – Present

- Engineered a hybrid-powered go-kart, integrating a Yamaha SRX 440 IC engine with four 12V electric motors for combined power delivery and regenerative braking
- Constructed the chassis using TIG welding (4130 chromoly steel), CNC machining (Al 6061 brackets), and ABS 3D printing (angled joints), reducing total weight by 15% versus a traditional welded-only frame

MUDDSUB — September 2023 – May 2024

- Designed a butterfly-shaped rotating marker dropper using SolidWorks, 3D printing, and mechanical tools for our Autonomous Underwater Vehicle (AUV) competing in the international RoboSub competition

MUDD AMATEUR ROCKETRY CLUB — September 2023 – May 2024

- Constructed a L-1 certified rocket, concurrently engaging in various rocket designs and simulations through OpenRocket

SEW-EURODRIVE, ENGINEERING MANAGEMENT INTERNSHIP — June 2022 – November 2022

- Increased automotive assembly line efficiency by 10% by implementing lean manufacturing principles (Kanban) and optimizing material flow using time-motion studies
- Collaborated with production managers to implement a real-time inventory tracking system, reducing stock shortages by 10% and ensuring JIT material availability

MACHINE LEARNING WORKSHOP (HMC) — September 2023 – November 2023

- Implemented practical applications of CNN, RNN, and transformers that utilize Python ecosystems and real-world datasets.
- Acquired knowledge of data manipulation using pandas, logistic regression, neural network fundamentals, fine-tuning techniques, along with practical applications on various datasets, including IMDb and CIFAR-10.

RESEARCH PROJECTS

CAR WHEEL DIMENSION OPTIMIZATION — September 2022 - March 2023

- Mathematically computed the optimal dimension of a car wheel travelling at 120kmh with reflection of real-life conditions
- Utilized advanced hydrodynamics and mechanics concepts, such as Reynolds number analysis, turbulence, rolling force

ACTIVITIES

CLASS OF 2027 PRESIDENT *(For HMC's class of 2027)* — October 2023 – May 2024

- Planned bonding events for the entire class, hosting and organizing outings to create an engaging and supportive community
- Frequently informed the class regarding campus events and first-year news through email and social media, representing the class in various university outlets such as in the student senate, school newspaper, and other events

WOMEN IN STEM *(Co-Leader and Head of Digital Media)* — August 2021 – May 2023

- Researched materials and software trends to publish daily technical advocacy content for Women in STEM on social media

VARSITY SWIMMER & SWIM COACH *(NCAA D3 athlete, Founder, and Team Captain)* — September 2010 – May 2024

- Founded free swimming lessons for underrepresented athletes willing to improve by hosting weekly practice sessions
- Led an inclusive, supporting, and competitive team with over 90 student athletes while also achieving 48 Gold, 34 Silver, and 37 Bronze medals from swimming throughout competitive career

SKILLS & AWARDS

Technical Skills: Fusion360, SolidWorks, MATLAB, ANSYS, Tolerance Analysis (DFM, DFA), CNC Machining, 3D Printing, Welding, Injection Molding, Python, Javascript, HTML, CSS, Unity, Blender

Languages: Fluent in Chinese, Korean, English, Limited Proficiency in Japanese

Certifications & Training: Google Project Management (Google, Coursera), Full Stack Engineering (Codecademy), Mechanical Engineering (Rice University, Coursera), Complete Blender Creator: 3D Modelling (GameDev.tv)

Awards: Davies Engineering Prize (2024), WES Global Economics Class Valedictorian (2020), HS Honors Award (2021)